RAJALAKSHMI ENGINEERING COLLEGE

Department of Artificial Intelligence and Machine Learning

AD23632-Framework for Data and Visual Analytics (FDVA)

MINI PROJECT REPORT

SOCIAL MEDIA ANALYTICS

Submitted in partial fulfilment of the requirements for the course
Framework for Data and Visual Analytics

Submitted by

Name	: SOWMYA R
Register Number	: 241501207
Department	: Artificial Intelligence and Machine Learning
College	: Rajalakshmi Engineering College

Submitted to

Faculty Name	: AKSHAYA V

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ABSTRACT

Social media platforms generate large amounts of data every day through user interactions such as likes, comments, shares, views, and follower growth. Analyzing this data helps understand user engagement, compare platform performance, and identify content that attracts more attention.

The project titled Social Media Analytics focuses on analyzing social media engagement data using Python and Microsoft Power BI. Python is used for data preprocessing, cleaning, exploratory data analysis, and statistical analysis. Power BI is used to create interactive visualizations that present important metrics and trends in an understandable format.

The dashboard includes key performance indicators such as total views, total likes, and total engagement. It also presents comparisons of engagement across platforms, engagement by post type, average engagement rate, follower distribution, engagement trends over time, and comparisons of comments and shares.

The project demonstrates how data analytics and visualization can transform raw social media data into meaningful insights that support better content planning and performance evaluation.

Keywords: Social Media Analytics, Python, Power BI, Data Visualization, Engagement Analysis, Exploratory Data Analysis.

CHAPTER 1: INTRODUCTION

1.1 Introduction

Social media has become an important communication and marketing platform for individuals, businesses, and organizations. Platforms such as YouTube, Instagram, Facebook, and LinkedIn allow users to publish content and interact with audiences.

The performance of social media content can be evaluated using various metrics, including views, likes, comments, shares, and followers. However, analyzing these metrics manually can be difficult when dealing with large datasets.

Social Media Analytics uses data analysis techniques to examine these metrics and understand audience engagement. Data visualization tools help represent the results through charts, graphs, and dashboards.

This project uses Python for data processing and exploratory analysis and Microsoft Power BI for visualizing the results. The dashboard enables users to compare platform performance and understand engagement patterns.

1.2 Problem Statement

Social media performance data contains multiple metrics that must be analyzed to understand audience behavior and content effectiveness. Examining individual records manually makes it difficult to identify trends and compare platforms.

Therefore, a simple analytical solution is required to process social media data, calculate important engagement metrics, and present the results through visualizations.

1.3 Objectives

The main objectives of the project are:

1. To analyze social media engagement data using Python.
2. To preprocess and clean the dataset.
3. To calculate total engagement and engagement rate.
4. To compare engagement across different social media platforms.
5. To analyze engagement by post type.
6. To visualize follower distribution across platforms.

7. To examine engagement trends over time.
8. To develop a Power BI dashboard for presenting analytical insights.

1.4 Scope of the Project

The project focuses on analyzing a sample social media dataset containing platform details, post information, and engagement metrics.

The scope includes:

- Data cleaning and preprocessing.
- Exploratory data analysis using Python.
- Calculation of engagement metrics.
- Comparison of different social media platforms.
- Visualization of engagement patterns.
- Development of a multi-page Power BI dashboard.

The analysis is limited to the data available in the selected dataset. The results do not necessarily represent the current performance of real social media accounts.

CHAPTER 2: TOOLS AND TECHNOLOGIES USED

2.1 Python

Python is a high-level programming language widely used for data analysis and visualization.

In this project, Python is used to inspect the dataset, identify missing values and duplicate records, process date information, calculate engagement metrics, perform statistical analysis, and generate charts.

Libraries Used

Pandas: Used for loading, cleaning, manipulating, and analyzing tabular data.

NumPy: Used for numerical operations and calculations.

Matplotlib: Used to create charts and graphs for visualizing the dataset.

2.2 Microsoft Power BI

Microsoft Power BI is a business intelligence and data visualization tool used to transform data into interactive reports and dashboards.

In this project, Power BI is used to:

- Import the social media dataset.
- Create calculated columns and aggregate metrics.
- Display key performance indicators.
- Compare engagement across platforms.
- Visualize post-type performance.
- Present follower distribution.
- Analyze trends over time.

2.3 CSV Dataset

The dataset is stored in Comma-Separated Values (CSV) format. This format stores tabular data in rows and columns and can be imported into Python and Power BI.

The dataset contains social media platform details, post information, and engagement-related metrics.

2.4 Technologies Summary

Table 2.1: Technologies and Their Purpose

Technology	Purpose
Python	Data preprocessing and analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Microsoft Power BI	Interactive dashboard development
CSV	Dataset storage
Google Colab	Running Python code in a browser, if used for the analysis

CHAPTER 3: DATASET DESCRIPTION

3.1 Dataset Overview

The project uses a structured dataset containing social media post performance information. Each row represents a record associated with a social media post.

The dataset enables comparisons between platforms and post types by examining views, likes, comments, shares, and followers. The following table describes the columns used in the project.

3.2 Dataset Attributes

Table 3.1: Dataset Attributes

Attribute	Description
Date	Date associated with the social media record
Platform	Social media platform, such as YouTube, Instagram, Facebook, or LinkedIn
Views	Number of views recorded for the post
Likes	Number of likes received
Comments	Number of comments received
Shares	Number of times the content was shared
Followers	Follower count associated with the platform or record
Post_ID	Unique or identifying post reference
Post_Type	Type of content, such as video, reel, short, image, story, or post
Total Engagement	Combined number of likes, comments, and shares
Engagement Rate	Engagement expressed as a percentage according to the formula used

Note: *Total Engagement and Engagement Rate may be calculated during preprocessing or in Power BI rather than being present in the original CSV file.*

3.3 Key Performance Metrics

Total Views

Total Views represents the sum of the view counts across the records included in the analysis.

Total Likes

Total Likes represents the sum of likes recorded across the dataset.

Total Engagement

Total Engagement combines three important interaction metrics:

Total Engagement = Likes + Comments + Shares

This metric provides a simple measure of the interactions received by social media content.

Engagement Rate

Engagement Rate expresses engagement relative to another metric, commonly views or followers. For this project, if the calculation is based on views, the formula is:

Engagement Rate (%) = (Total Engagement ÷ Views) × 100

The formula should match the one implemented in the final dataset or Power BI report.

CHAPTER 4: METHODOLOGY

4.1 Project Workflow

The project follows these steps:

1. Dataset collection and loading.
2. Data inspection.
3. Data cleaning and preprocessing.
4. Exploratory data analysis using Python.
5. Calculation of engagement metrics.
6. Data visualization and comparison.
7. Dashboard creation using Power BI.
8. Interpretation of analytical results.

Workflow:

Dataset → Data Cleaning → Data Analysis Using Python → Visualization Using Power BI → Insights and Conclusions

4.2 Data Loading

The CSV dataset is loaded into Python using the Pandas library.

The dataset is inspected to understand its dimensions, column names, data types, and initial records. This helps identify the structure of the data before performing further operations.

4.3 Data Cleaning and Preprocessing

Data cleaning is an important step because missing or duplicate records and incorrect data types may affect the accuracy of the analysis.

The preprocessing stage includes:

- Checking the number of rows and columns.
- Inspecting column names and data types.
- Identifying missing values.
- Checking duplicate records.
- Converting the date column into an appropriate date format.
- Creating calculated engagement metrics when required.

- Preparing the cleaned dataset for analysis.

Any missing values or duplicate records are handled according to the preprocessing logic used in the Python program.

4.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the dataset and identify patterns.

The analysis includes:

- Calculating descriptive statistics.
- Examining views, likes, comments, and shares.
- Grouping engagement by platform.
- Comparing engagement by post type.
- Examining engagement over time.
- Comparing follower counts across platforms.

The results help identify differences in engagement across the available categories.

4.5 Dashboard Development

The processed data is imported into Microsoft Power BI to develop a four-page dashboard.

The dashboard uses KPI cards, column charts, a donut chart, a treemap, a line chart, and a combination chart to present the results. Each page focuses on a different aspect of social media performance.

CHAPTER 5: POWER BI DASHBOARD AND VISUALIZATIONS

5.1 Dashboard Overview

The Power BI report contains four pages.

The first page presents overall performance metrics and engagement trends. The second page compares engagement by platform and post type. The third page examines engagement rates and follower distribution. The fourth page compares comments and shares across platforms.

Together, these pages provide a visual summary of the dataset.

5.2 Page 1: Overall Performance and Engagement Trend

Visualizations used:

- KPI card for total views.
- KPI card for total likes.
- KPI card for total engagement.
- Line chart showing total engagement by date.

KPI Cards

The first three cards summarize the overall performance of the dataset. Based on the screenshot, the displayed values are approximately:

Table 5.1: KPI Values

KPI	Displayed Value
Total Views	3M
Total Likes	265K
Total Engagement	307K

Note: *These are rounded display values. The precise values can be obtained from Power BI.*

Line Chart: Total Engagement by Date

The line chart shows how total engagement varies over time. The horizontal axis represents dates, while the vertical axis represents total engagement.

Purpose:

- Identify changes in engagement over time.

- Observe periods with higher or lower engagement.
- Identify unusual peaks in the dataset.

Observation: The displayed line chart shows fluctuations in engagement over the selected period, with a noticeable increase near the end of the timeline. This indicates that engagement varies over time within the dataset. Further investigation would be required to determine the reasons for these changes.

5.3 Page 2: Platform and Post-Type Analysis

Visualizations used:

- Column chart for total engagement by platform.
- Donut chart for total engagement by post type.

Column Chart: Total Engagement by Platform

This chart compares the total engagement recorded for each social media platform. The platforms shown are YouTube, Instagram, Facebook, and LinkedIn. Based on the screenshot, the approximate engagement values are:

Table 5.2: Approximate Total Engagement by Platform

Platform	Approximate Total Engagement
YouTube	138K
Instagram	99K
Facebook	49K
LinkedIn	21K

Note: *These values are approximate readings from the chart and should be verified in Power BI before being included as exact results.*

Observation: YouTube has the highest total engagement in the displayed chart, followed by Instagram, Facebook, and LinkedIn. This indicates that YouTube contributes the largest share of total engagement in this particular dataset. However, this comparison alone does not establish that YouTube always performs better than the other platforms.

Donut Chart: Total Engagement by Post Type

The donut chart displays how total engagement is distributed among different content types. The categories shown include:

- Video

- Reel
- Short
- Image
- Story
- Post

Observation: The donut chart shows differences in engagement among the content types. Video and reel content account for substantial portions of the total engagement displayed. This visualization helps compare the relative contribution of different post types.

5.4 Page 3: Engagement Rate and Follower Distribution

Visualizations used:

- Column chart for average engagement rate by platform.
- Treemap for followers by platform.

Column Chart: Average Engagement Rate by Platform

This chart compares the average engagement rate of the four platforms. The vertical axis represents the average engagement rate, and the horizontal axis lists the platforms.

Observation: Based on the screenshot, Instagram has the highest average engagement rate among the displayed platforms, followed by YouTube, LinkedIn, and Facebook. This suggests that Instagram performs well in terms of the engagement rate calculated for this dataset.

Important distinction: The platform with the highest total engagement is not necessarily the platform with the highest engagement rate. Total engagement measures the volume of interactions, whereas engagement rate measures interactions relative to the denominator used in the calculation.

Treemap: Followers by Platform

A treemap represents values using rectangles of different sizes. In this dashboard, the size of each rectangle represents the follower count associated with a platform.

Observation: YouTube occupies the largest area in the treemap, followed by Facebook, Instagram, and LinkedIn. This indicates that YouTube has the highest aggregated follower count among the platforms in the displayed dataset.

5.5 Page 4: Comments and Shares by Platform

Visualization used: Combination chart containing columns and a line.

The chart compares the sum of comments and shares for YouTube, Instagram, Facebook, and LinkedIn. The columns represent the sum of comments, while the line represents the sum of shares.

Purpose

This visualization helps compare two interaction metrics across multiple platforms.

Observation: YouTube records the highest number of comments in the displayed chart, followed by Instagram, Facebook, and LinkedIn. The line representing shares also decreases across the displayed platforms, with YouTube showing the highest value and LinkedIn the lowest. This suggests that YouTube contributes the largest number of comments and shares among the platforms in the current dataset.

CHAPTER 6: RESULTS AND DISCUSSION

6.1 Results

The Social Media Analytics project successfully presents a multi-page dashboard using Microsoft Power BI. The report summarizes key performance indicators and provides comparisons across social media platforms, post types, and dates.

The main findings from the displayed dashboard are:

Overall performance: The dashboard displays approximately 3 million views, 265 thousand likes, and 307 thousand total engagements.

Platform comparison: YouTube has the highest total engagement among the four displayed platforms.

Post-type comparison: Engagement differs across videos, reels, shorts, images, stories, and posts.

Engagement rate: Instagram has the highest average engagement rate in the displayed chart.

Follower distribution: YouTube has the highest aggregated follower count in the treemap.

Comments and shares: YouTube records the highest values for both metrics in the displayed combination chart.

Engagement trends: The line chart shows fluctuations over time, including a noticeable increase near the end of the displayed period.

These findings are based on the values and patterns visible in the submitted dashboard screenshots.

6.2 Discussion

The dashboard demonstrates how social media performance can be evaluated using multiple metrics instead of relying on a single indicator.

For example, YouTube leads in total engagement, while Instagram leads in average engagement rate. This illustrates the importance of examining both the volume of interactions and the engagement rate.

Similarly, the post-type chart helps identify differences between content categories, while the line chart provides a time-based view of engagement.

The visualizations make the data easier to interpret and allow users to identify patterns without manually examining every record.

The analysis is descriptive and does not establish why a particular platform or post type performs better. Additional information, such as audience demographics, posting frequency, and content quality, would be needed to investigate those factors.

CHAPTER 7: ADVANTAGES AND LIMITATIONS

7.1 Advantages

- Provides a visual summary of social media performance.
- Simplifies the comparison of engagement across platforms.
- Helps identify differences between content types.
- Presents important metrics through KPI cards.
- Shows engagement patterns over time.
- Combines Python-based analysis with Power BI visualization.
- Makes complex tabular data easier to understand.

7.2 Limitations

- The analysis depends on the completeness and accuracy of the selected dataset.
- The results may not represent real-time social media performance.
- The dataset may not include audience demographics or other factors influencing engagement.
- The dashboard describes patterns but does not establish the causes of those patterns.
- The conclusions are limited to the platforms and metrics available in the dataset.

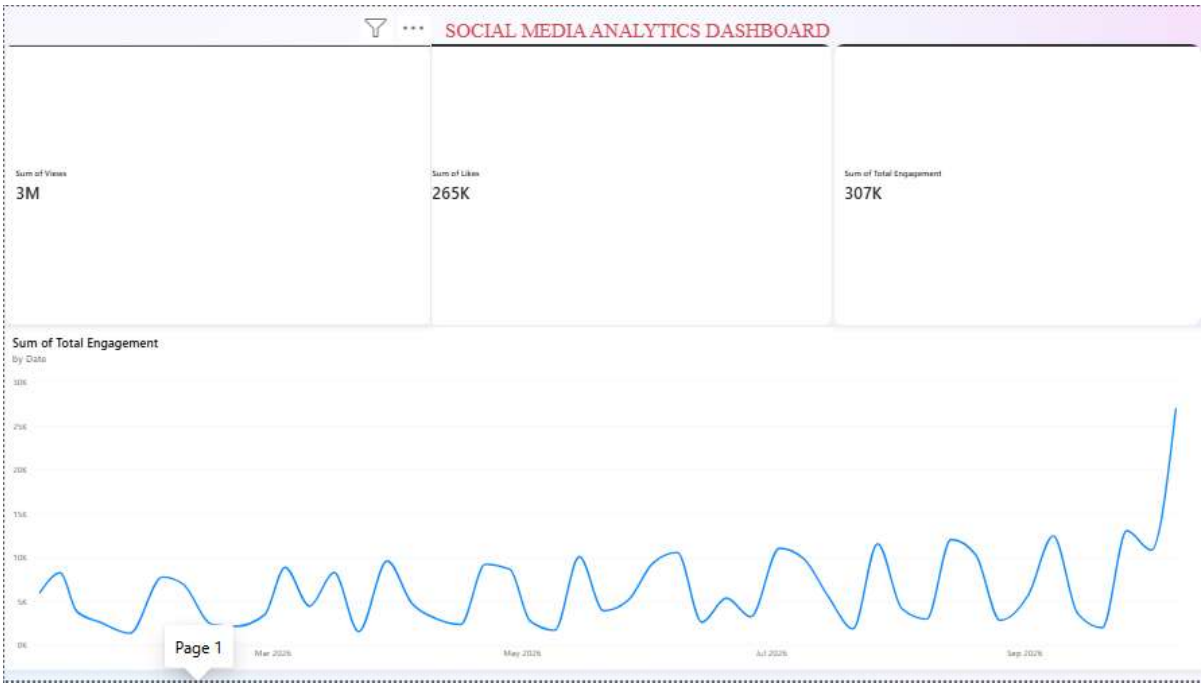
7.3 Future Enhancements

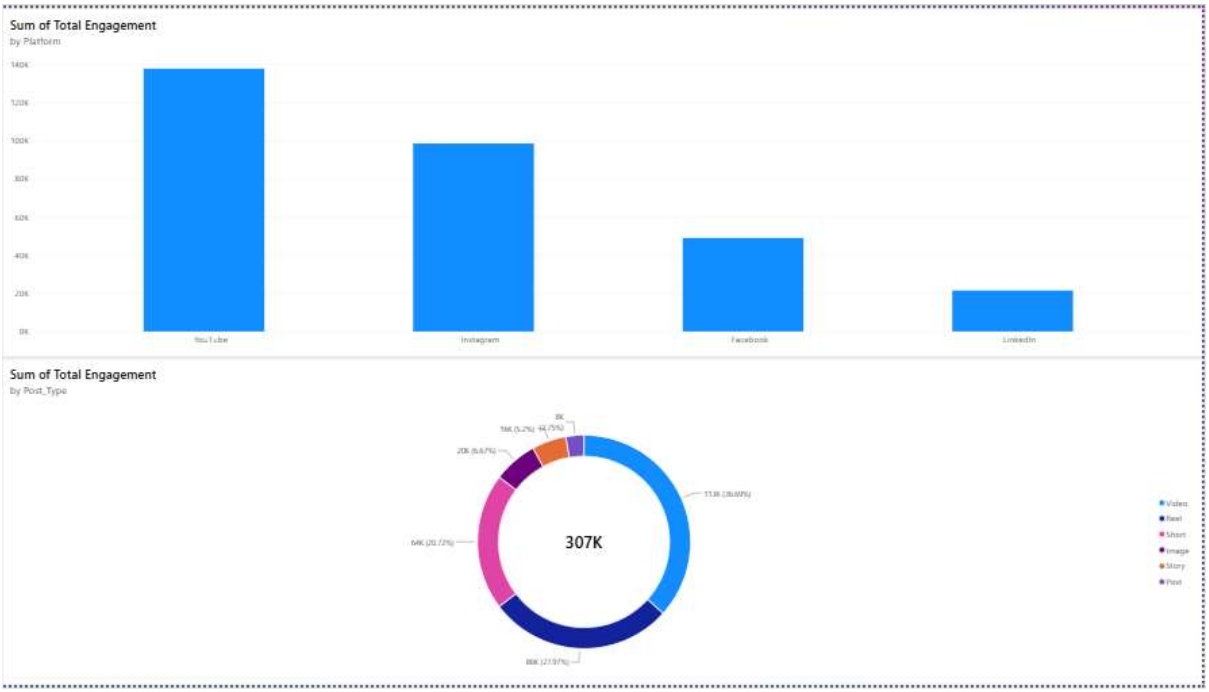
The project can be improved in the following ways:

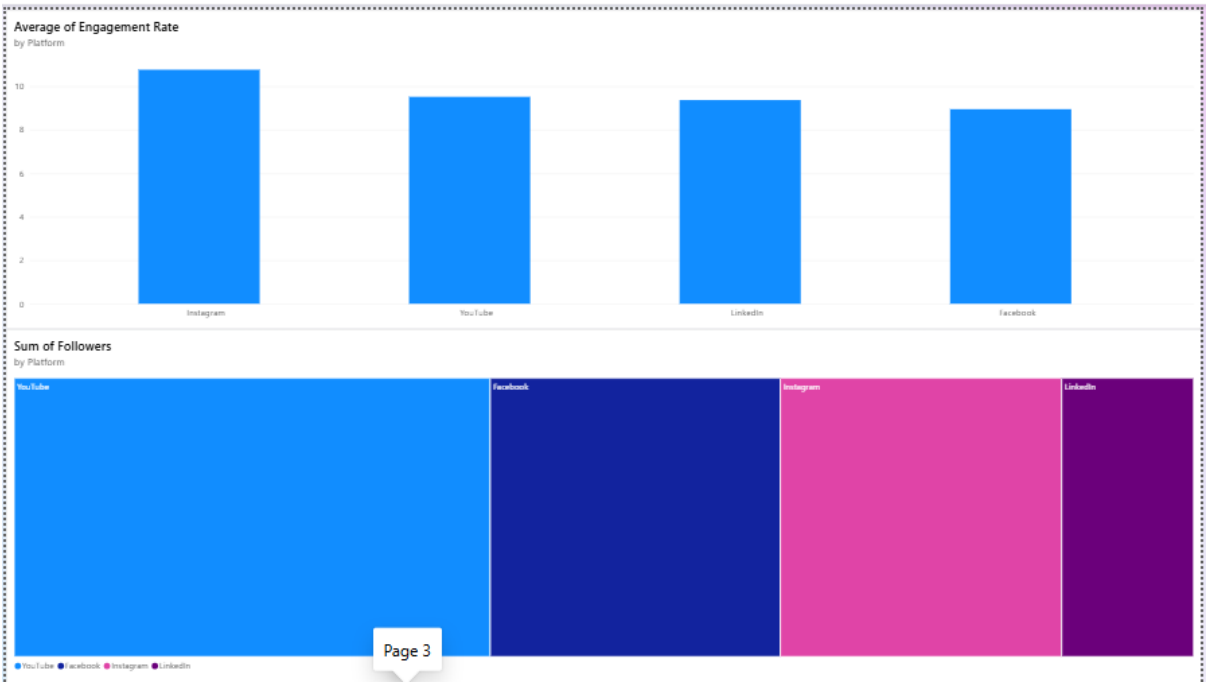
- Connect the dashboard to regularly updated data.
- Include additional social media platforms.
- Add interactive filters for platform, date, and post type.
- Include sentiment analysis of comments.
- Use machine learning to predict engagement.
- Identify the best-performing posting times.
- Add more detailed audience and content analysis.

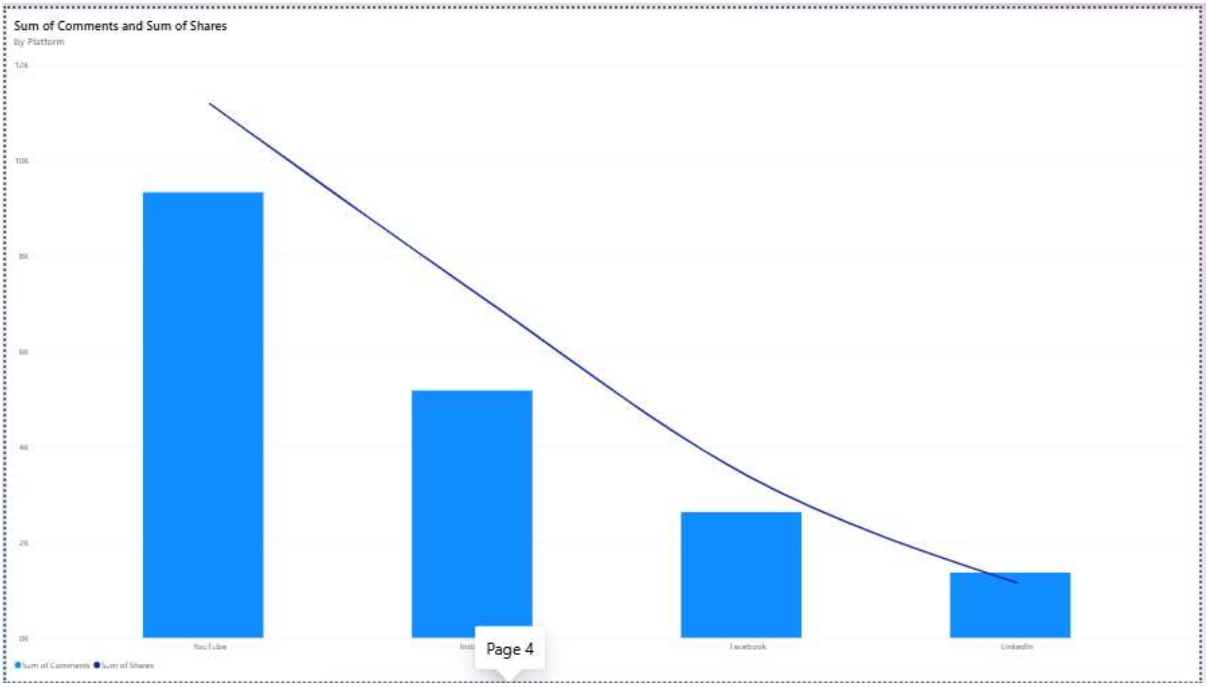
These enhancements could make the project more useful for social media managers and digital marketing teams.

CHAPTER 8: POWER BI DASHBOARD OUTPUT






Average of Engagement Rate
by Platform
10
8
6
4
2
0
Instagram
YouTube
LinkedIn
Facebook
Sum of Followers
by Platform
YouTube
Facebook
Instagram
LinkedIn
Page 3
YouTube Facebook Instagram LinkedIn


Sum of Comments and Sum of Shares
by Platform
12K
10K
8K
6K
4K
2K
0K
YouTube
Facebook
LinkedIn
Page 4
Sum of Comments
Sum of Shares

CHAPTER 9: CONCLUSION

The Social Media Analytics project demonstrates the application of Python and Microsoft Power BI in analyzing and visualizing social media performance data.

Python supports data inspection, preprocessing, and exploratory data analysis, while Power BI presents the results through KPI cards and visualizations.

The dashboard enables comparisons of total engagement across platforms, engagement by post type, average engagement rates, follower distribution, engagement trends, and comments and shares.

The analysis shows that different platforms may perform differently depending on the metric being examined. Therefore, evaluating multiple metrics provides a more comprehensive understanding of social media performance.

Overall, the project provides practical experience in data preprocessing, exploratory analysis, data visualization, and dashboard development. It demonstrates how raw data can be converted into meaningful visual information that supports better understanding and decision-making.